

Introduction to Macroeconomics (S102016CR)

Spring Semester

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graph TD; A[Fourth Part: Macroeconomic Policies] --> B[AD and AS model (Chapter 10)]; A --> C[Government Policies (Chapter 11)]; A --> D[Inflation and Unemployment (Chapter 12)];
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Fourth Part:
Macroeconomic
Policies

AD and AS
model
(Chapter 10)

**Government
Policies
(Chapter 11)**

Inflation and
Unemployment
(Chapter 12)

Week 11:
Impact of Monetary and
Fiscal Policy on
Aggregate Demand

In the previous session, we saw how income and unemployment could deviate from their full employment levels in the short run.

The long run is made of a series of short runs. Understanding how the government can minimize these deviations around the full employment equilibrium is important, as these deviations can be very costly.

In particular, we will explore the role that Monetary and Fiscal policies can play in minimizing these deviations from full employment.

These were at the core of John Maynard Keynes's book: *The General Theory of Employment, Interests, and Money*, published in 1935.

Neo-classical economists believe that in the long run, we will return to full employment, as we saw in the earlier session, so these interventions are unnecessary.

Real Business Cycles economists and economists from the Austrian School (Hayek) also believe that they create the wrong incentives and prevent the economy from retrieving a healthier equilibrium.

Thus, this session will end with a discussion of the pros and cons of government intervention (and a bit of rap)

Chapter Contents

- a) Impact of Monetary and Fiscal Policies on AD
- b) The Keynesian Multiplier
- c) The Crowding-Out Effect
- d) Pros and Cons of Stabilization Policies
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Chapter Contents

a) Impact of Monetary and Fiscal Policies on AD

b) The Keynesian Multiplier

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d) Pros and Cons of Stabilization Policies

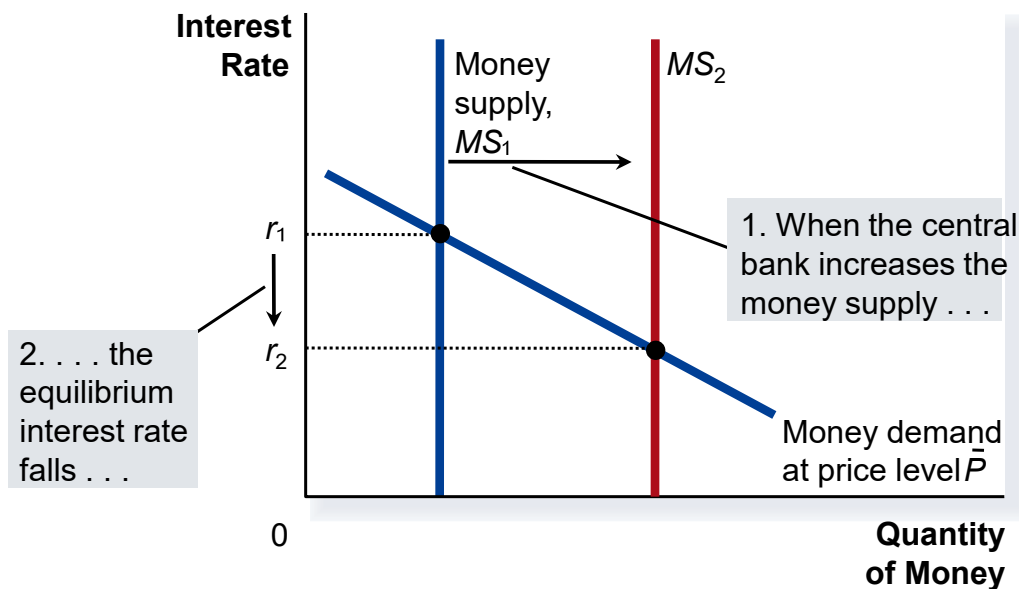
e) Summary

a. Impact of Monetary and Fiscal Policies on AD

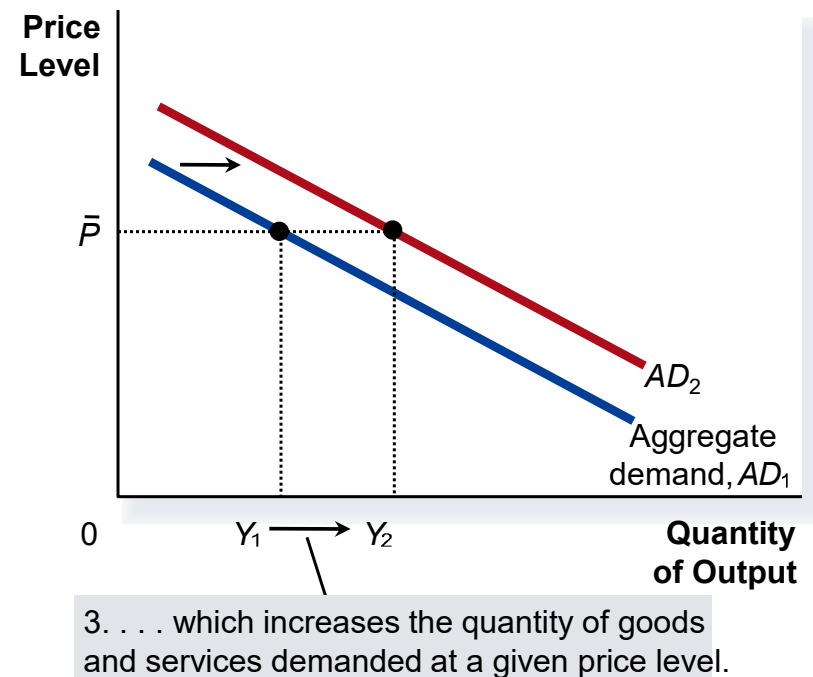
- Many factors influence aggregate demand besides monetary and fiscal policy. In particular, desired spending by households and business firms.
- When desired spending changes, aggregate demand shifts, causing short-run fluctuations in output and employment. Monetary and fiscal policy sometimes offset those shifts and stabilize the economy.

Monetary Policy

(a) The Money Market



(b) The Aggregate Demand Curve



But do Central Banks conduct Monetary Policy in terms of quantity of money or interest rate?

It doesn't matter (see the discussion in chapter 6 for how CB conducts monetary policy).

Because changes in the money supply lead to changes in interest rates, monetary policy can be described either in terms of the money supply or the interest rate. They will have the same impact on investment, and therefore on aggregate demand

In recent years, many of the major central banks have conducted policy by setting the interest rate at which they will lend to the banking sector (the repo rate for the Swiss National Bank or the Bank of England, the refinancing rate for the ECB, and the discount rate in the terms of the US Federal Reserve) and not in terms of quantity of money.

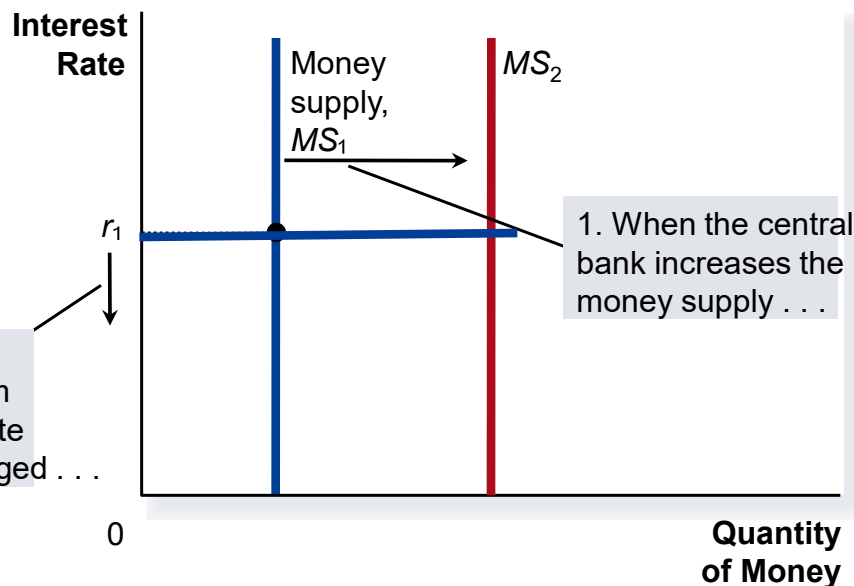
But quantitative easing policies have changed that

A **liquidity trap** is a situation in which Monetary Policy is not effective anymore (individuals keep all cash under the mattress) because of a lack of trust in the banking system.

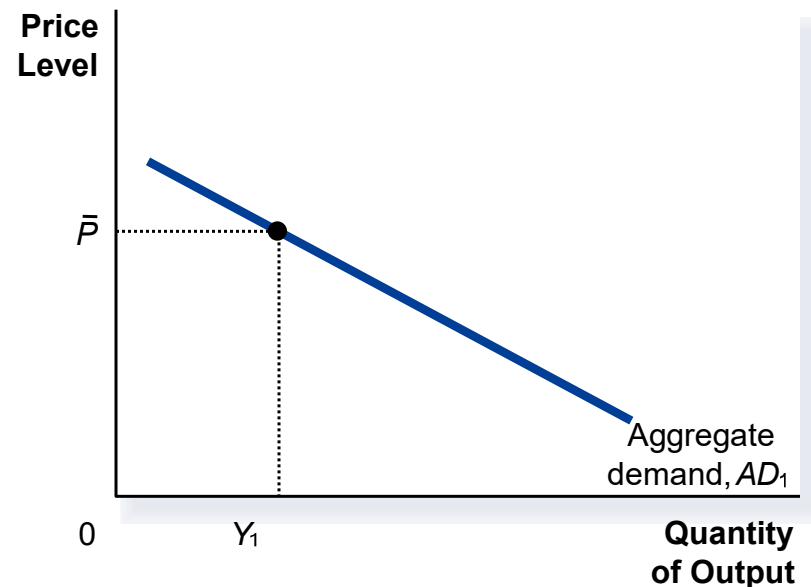
The CB can increase the amount of money in circulation without this having any impact on the interest rate or it can decrease the interest rate without having any impact on money demand. All money is already in cash.

Under a **liquidity trap**, monetary policy has no impact on investment or AD

(a) The Money Market



(b) The Aggregate Demand Curve



Fiscal Policy

- The government conducts fiscal policy. It refers to the government's choices over the level of government purchases (G) or taxes (T).
- When policymakers increase taxes, aggregate demand falls because consumers reduce spending.
- When policymakers decrease taxes, aggregate demand increases because consumers can increase spending.
- When the government alters its purchases of goods or services, it directly shifts the aggregate demand curve: $AD = C + I + G + X - M$.
- If G increases, AD increases. If G falls, AD falls.

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b. The Keynesian Multiplier

- Government purchases are said to have a *multiplier effect* on aggregate demand.
 - Each pound spent by the government can raise the aggregate demand for goods and services by more than a pound.
- The multiplier effect refers to the additional shifts in aggregate demand that result when expansionary fiscal policy increases income and thereby increases consumer spending.
- It is called a multiplier because the additional increase in government expenditure increases income, which increases consumption, increasing income, and therefore consumption again, etc....
- The formula for the multiplier is:
$$\text{Multiplier} = 1/(1 - \text{MPC})$$
$$\Rightarrow \Delta Y = 1/(1 - \text{MPC}) \Delta G$$
- An important number in this formula is the marginal propensity to consume (*MPC*). It is the fraction of extra income that a household consumes rather than saves.

Example

- Suppose that if we are given an extra dollar our consumption will increase by 75 cents and our savings by 25 cents.

- Then the *MPC* is $3/4$, and the Keynesian Multiplier will be:

$$\text{Keynesian Multiplier} = 1/(1 - 3/4) = 4$$

- In this case, a £10 billion increase in government spending generates £40 billion of increased demand for goods and services.

- Note that the Keynesian multiplier will also work if other determinants of income increase exogenously (i.e., Investment or Exports).
- For example, if the ROW experiences a boom and imports more of our goods, then the increase in domestic income will be larger than the increase in exports. The Keynesian multiplier will give an increase in domestic income
- It will also be present if investment increases because of a shift in the money supply that leads to an increase in investment.

The Keynesian multiplier of lump-sum taxes

- Expansionary fiscal policies include tax reductions (not only increases in government expenditure). If taxes are lump-sum, the multiplier is:
Lump-sum taxes Multiplier = $- MPC/(1-MPC)$
 $\Rightarrow \Delta Y = -MPC/(1-MPC) \Delta T$
- The lump-sum tax multiplier is smaller than the government expenditure multiplier.

The balanced budget Keynesian multiplier

- Because the lump-sum tax Keynesian multiplier is smaller than the government expenditure multiplier, it is possible to increase real income while increasing both lump-sum taxes and government expenditure
- The balanced budget multiplier is given by:
Balanced budget multiplier = 1
 $\Rightarrow \Delta Y = \Delta G = \Delta T$

The Keynesian multiplier with income taxes

- In an economy with income taxes, the multiplier will be different because as income increases, so do taxes, and therefore consumption may not increase by as much. There is a leak in the multiplier, which will be smaller.
Keynesian Multiplier with income taxes = $1/(1 - MPC*[1-t])$, where t is the income tax rate.
- The higher is t , the lower the multiplier.
- It implies: $\Delta Y = 1/(1-MPC*[1-t]) \Delta G$

The Keynesian multiplier in an open economy

- In an open economy where imports depend on domestic income, the multiplier will be different because as income increases, imports increase, which does not boost demand for domestically produced goods. There is a leak in the multiplier. It will be smaller:
Open Economy Keynesian Multiplier = $1/(1 - MPC + MPI)$,
where MPI is the marginal propensity to import.
- The higher is MPI the lower the multiplier
- It implies $\Delta Y = 1/(1-MPC+MPI) \Delta G$

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c. The Crowding-Out Effect

- Fiscal policy may not affect the economy as strongly as predicted by the multiplier.
- An increase in government purchases leads to a higher government deficit (lower government savings) which causes the interest rate to rise (the market for loanable funds). A higher interest rate results in lower investment spending.
- This results in a reduction in aggregate demand. This effect is called the *crowding-out effect*.
- The crowding-out effect dampens the effects of fiscal policy on aggregate demand.
- The final impact on Y will be larger than the increase in G only if the multiplier effect is larger than the crowding-out effect.

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d. Pros and Cons of Stabilization Policies

- Economic stabilization has been an explicit goal of government policy in West European and North American countries since 1945.

Pros:

- Short-run fluctuations can leave an economy well below its full employment equilibrium for extended periods (1929, 2008, 2020). This is inefficient but also has strong social consequences.
- Governments try to prevent this through an active fiscal and monetary policy
- The lack of government action in the 1930s (influenced by the view of neo-classical economists) probably led to a longer-than-necessary crisis, according to the view of John Maynard Keynes.
- Crisis can also be self-fulfilling. If everyone believes there will be a crisis, we will stop consuming and start saving, and the crisis will occur.
- A strong and credible fiscal or monetary policy can reverse this self-fulfilling pessimism.

Cons:

- Monetary and fiscal policy affect the economy with a substantial lag:
 - Investment is often determined many years in advance, and monetary policy will affect investment too late.
 - Fiscal Policy needs the agreement of many political actors. By the time we finish negotiating, the crisis is over
- There is crowding-out of private investment
- And Ricardian equivalence tells us fiscal policies will have no impact anyway.
- There are anyway *automatic stabilizers* that stimulate aggregate demand when the economy goes into a recession without policymakers having to take any deliberate action.
 - As income falls, income taxes will fall, and social spending (unemployment benefits) by the government will increase.
- And markets need a correction when bad decisions have been made: e.g., market bubbles promoted by cheap money for example (Hayek's views)
- Who is right? Your call...Here is some more Keynes-Hayek rap: [Keynes vs Hayek second round](#)

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e. Summary

- Keynes proposed the liquidity preference theory to explain the interest rate's determinants.
- According to this theory, the interest rate adjusts to balance the supply and demand for money.
- An increase in the price level raises money demand and increases the interest rate.
- A higher interest rate reduces investment and, thereby, the quantity of goods and services demanded.
- The downward-sloping aggregate demand curve expresses this negative relationship between the price level and the quantity demanded.
- Policy makers can influence aggregate demand with monetary policy.
- An increase in the money supply will ultimately shift the aggregate demand curve to the right.
- A decrease in the money supply will ultimately shift the aggregate demand curve to the left.

- When the government alters spending or taxes, the resulting shift in aggregate demand can be larger or smaller than the fiscal change.
- The multiplier effect tends to amplify the effects of fiscal policy on aggregate demand.
- The crowding-out effect tends to dampen the effects of fiscal policy on aggregate demand.
- Because monetary and fiscal policy can influence aggregate demand, the government sometimes uses these policy instruments to stabilize the economy.
- Economists disagree about how active the government should be in this effort.
 - Advocates say that if the government does not respond, the result will be undesirable fluctuations.
 - Critics argue that stabilization attempts often become destabilizing in the long run because they create the wrong incentives.